

CHAPTER-01**ABSTRACT**

Maternal health is one of the most critical aspects of public health, and ensuring the safety and well-being of pregnant women is essential for both the mother and the developing child. Pregnancy involves significant physiological changes, and any abnormality in vital signs can indicate potential complications. Timely intervention during emergencies can make a life-saving difference, especially in remote or underserved regions where access to healthcare may be limited. To address these challenges, the proposed project focuses on a **Safety Band for Prenatal Care**, an innovative wearable device designed to enhance maternal safety using IoT technology. This smart wearable integrates continuous health monitoring, real-time location tracking, and instant emergency communication, providing a comprehensive solution for prenatal care. The core of the system is the **ESP32 microcontroller**, a versatile device with built-in Wi-Fi and Bluetooth capabilities that allows seamless integration with multiple sensors and communication modules. The ESP32 acts as the central processing unit, continuously collecting data from attached sensors, processing it, and sending necessary alerts in real time. The device includes a **pulse sensor** to monitor heart rate and a **temperature sensor** to track body temperature. These vital health parameters are continuously measured, enabling the early detection of abnormalities such as fever, stress, or unusual heart rates. Such real-time monitoring ensures that potential risks are identified before they escalate into severe complications, offering peace of mind to both the expectant mother and her family. In addition to health monitoring, the safety band incorporates a **GPS module** that provides real-time location tracking. This feature is particularly valuable during emergencies, as healthcare providers or family members can quickly locate the user and dispatch assistance without delay. The combination of health data and location information makes the device especially effective in scenarios where the user may be traveling alone, living in a remote area, or unable to call for help immediately. Furthermore, the device is equipped with an **SOS button**, which, when pressed, triggers an automatic alert system. The ESP32 sends a message via **Telegram**, a popular messaging platform, containing the user's current health parameters and live GPS coordinates. This instant communication channel ensures that emergency responders or family members are informed immediately, reducing response times and potentially saving lives.

The user interface of the safety band is designed for simplicity and convenience. An **OLED display** provides immediate feedback to the wearer, showing vital signs, battery status, and confirmation of alerts sent. This ensures that the user remains aware of her health status and the device's functionality without needing additional tools or devices. To ensure portability and comfort, the device is powered by **rechargeable lithium cells**, managed through a **step-down voltage regulator** to safely distribute power to the microcontroller and sensors. The compact design of the battery holder, along with the lightweight, belt-style wearable form factor, allows the device to be worn throughout the day without causing discomfort or hindrance to daily activities.

CHAPTER-02

INTRODUCTION

Maternal health stands as a paramount indicator of a nation's healthcare system and societal well-being. The period of pregnancy, while a natural and joyous journey, is a time of profound physiological change that carries inherent risks for both the mother and the developing child. The global healthcare community, through initiatives like the United Nations' Sustainable Development Goals, has long strived to reduce preventable maternal mortality. Despite significant progress, the stark reality is that hundreds of thousands of women still succumb to complications during pregnancy and childbirth each year. A vast majority of these tragic outcomes are preventable and are disproportionately concentrated in remote, rural, and underserved regions where access to timely and effective medical care is severely limited. The conventional model of prenatal care, which relies on scheduled, periodic visits to a clinic, creates significant gaps in monitoring. While these check-ups are essential, they only provide a snapshot of health at a single point in time, leaving the expectant mother unmonitored for long intervals. It is within these intervals that acute, life-threatening complications such as the sudden onset of preeclampsia, gestational hypertension, cardiac distress, or severe infections leading to sepsis can develop and escalate rapidly. For a woman living miles away from the nearest medical facility, with limited transportation and communication infrastructure, recognizing the onset of an emergency and successfully summoning help is a monumental challenge. This delay in detection and response is often the critical factor that leads to preventable disability or death. The problem is thus twofold: a lack of continuous health oversight and the absence of a reliable, immediate channel for emergency communication.

To address these critical deficiencies, this project proposes the **Safety Band for Prenatal Care**, an innovative wearable device that leverages the power of the Internet of Things (IoT) to create a continuous safety net for expectant mothers. By shifting the paradigm from reactive, intermittent care to a proactive, constant monitoring system, this device aims to bridge the gap between the patient and healthcare provider. The solution is an integrated system designed for comfort, ease of use, and reliability, providing real-time data and an instantaneous emergency alert mechanism. It is conceived as a practical, affordable, and scalable tool to bring advanced monitoring capabilities directly to the individuals who need them most, regardless of their geographical location. The core of the Safety Band is a powerful yet low-cost **ESP32 microcontroller**, which serves as the central processing unit. This device is engineered to continuously collect data from a suite of integrated sensors, including a **pulse sensor** to monitor heart rate and a **temperature sensor** to track body temperature. These vital signs are fundamental indicators of maternal health and can provide early warnings of potential complications. In parallel, a **GPS module** is integrated to provide precise, real-time location tracking, a feature that is indispensable during an emergency. The user interface is kept simple and intuitive, with an **OLED display** providing immediate feedback on vital statistics and device status. The most crucial feature is a dedicated **SOS button**. When activated by the user, the system automatically captures the latest health data and

GPS coordinates and transmits them as an emergency alert via the **Telegram** messaging platform to a pre-configured list of contacts, such as family members, community health workers, or emergency services. The entire system is housed in a lightweight, belt-style form factor and powered by rechargeable lithium cells, ensuring portability and suitability for all-day wear.

The primary objective of this project is to design, develop, and validate a functional prototype of this life-saving device. This entails the successful integration of all hardware components, the development of robust firmware to manage data collection and communication, and the creation of a user-friendly physical design. The underlying motivation is profoundly humanitarian: to leverage accessible technology to reduce maternal mortality rates, empower pregnant women with a greater sense of security, and promote healthcare equity. The Safety Band for Prenatal Care is more than a technological gadget; it is envisioned as a vital tool that can provide peace of mind to families and offer a tangible layer of protection for the vulnerable journey of pregnancy, ensuring that help is always just a button press away.

2.1 OVERVIEW

The **Safety Band for Prenatal Care** is a comprehensive, wearable Internet of Things (IoT) system engineered to provide a robust safety net for expectant mothers through continuous health monitoring and an immediate emergency communication channel. The fundamental purpose of this project is to address critical vulnerabilities in modern prenatal care, particularly the lack of real-time physiological data between clinical appointments. By offering persistent monitoring of vital signs and precise location tracking, the device functions as a proactive guardian, designed to provide early warnings of potential complications and to drastically reduce response times during acute medical emergencies. The system is specifically conceptualized for two key user groups: pregnant women in remote or underserved areas with limited access to healthcare, and those managing high-risk pregnancies (such as cases involving preeclampsia, gestational diabetes, or cardiac conditions) who require a higher level of vigilance. It fundamentally shifts the care model from being reactive to proactive, empowering both the user and their support network with actionable, real-time information.

2.1.1 System Architecture and Core Components

The technological foundation of the Safety Band is a carefully integrated ecosystem of hardware chosen for its reliability, low power consumption, and cost-effectiveness.

- **Central Processing Unit:** The device's operation is managed by the **ESP32 microcontroller**. This component was selected for its powerful dual-core architecture, which allows for efficient multitasking for instance, one core can be dedicated to managing sensor data acquisition while the other handles Wi-Fi connectivity and communication protocols. Its integrated Wi-Fi and Bluetooth capabilities eliminate the need for additional modules, simplifying the design and reducing both size and cost.
- **Advanced Health Monitoring Sensors:** The band is equipped with a suite of non-invasive sensors to capture a holistic view of the mother's physiological state:

1. **Pulse Oximeter Sensor (MAX30100):** This sophisticated sensor provides two critical metrics: heart rate (pulse). Continuous heart rate monitoring can detect tachycardia or bradycardia, which can be indicators of stress or underlying cardiac issues. Simultaneously, tracking SpO2 is vital, as a significant drop can indicate respiratory or circulatory distress, a serious concern during pregnancy.
2. **Digital Temperature Sensor (DS18B20):** This sensor provides high-accuracy body temperature readings. Its primary function is the early detection of fever, a key symptom of maternal infections (such as chorioamnionitis) which, if left untreated, can pose severe risks to both mother and fetus.
- **Location, Communication, and User Interface:**
 1. **GPS Module (NEO-6M):** To ensure a swift emergency response, the band incorporates a GPS module that provides precise, real-time latitude and longitude coordinates. This eliminates any guesswork for responders trying to locate the user in an emergency.
 2. **Communication Protocol:** The ESP32's onboard Wi-Fi module is used to connect the device to the internet via a pre-configured network. For the alert mechanism, the system leverages the **Telegram Bot API**, a secure and reliable platform capable of instantly pushing notifications to individuals or groups.
 3. **OLED Display:** A high-contrast, energy-efficient Organic Light Emitting Diode (OLED) screen serves as the primary user interface. It displays vital signs, battery level, and system status confirmations, ensuring the user is always informed.
 4. **SOS Button:** A tactile, physical button is used as the dedicated emergency trigger. This design choice prioritizes reliability and ease of use under duress over more complex interfaces like touchscreens.
 5. **Power Management System:** The device is powered by a rechargeable **Lithium-Ion (Li-ion) battery**, managed by a comprehensive Battery Management System (BMS). This subsystem not only facilitates safe charging via a USB-C port but also provides critical overcharge and over-discharge protection, ensuring the safety and longevity of the power source.

2.1.2 Functional Workflow and Operational Logic

The device operates on a clear, event-driven workflow. In its **standard monitoring mode**, the firmware directs the ESP32 to periodically collect and analyze sensor data. These readings are displayed on the OLED screen and can be logged to create a health baseline over time. The system can be programmed with specific thresholds to trigger non-emergency visual alerts for the user. When the user activates the **emergency alert mode** by pressing the SOS button, a predefined, automated protocol is executed. The firmware captures a final, immediate snapshot of all sensor data, retrieves the latest GPS coordinates, and formats this information into a clear, data-rich alert message. This message containing the user's identity, vital signs, a timestamp, and a direct Google Maps link to their location is then transmitted instantly via the Telegram API to their designated emergency contacts. This entire process is completed within seconds, ensuring that help is dispatched with the maximum possible context and speed.

2.1.3 Design Philosophy and User Experience

The physical design of the Safety Band is guided by a philosophy of comfort, simplicity, and durability. The **ergonomic, belt-style form factor** ensures the device remains securely and comfortably in place, which is crucial for obtaining accurate sensor readings. The enclosure is designed to be lightweight and non-intrusive, while the strap is made from hypoallergenic materials suitable for continuous, long-term wear. The user experience is intentionally simplified to its essential functions to prevent any confusion, especially during a high-stress emergency. By building the system with accessible, off-the-shelf components, the project remains highly cost-effective, positioning it as a scalable solution for large-scale deployment in public health programs and low-resource environments. This thoughtful combination of advanced technology and user-centric design makes the Safety Band for Prenatal Care a powerful tool in the global effort to make pregnancy safer for all women.

2.2 OBJECTIVES

1. **Ergonomic Design:** Design a lightweight and hypoallergenic wearable band that is comfortable enough for continuous 24/7 use.
2. **Durable Enclosure:** Engineer a compact and robust enclosure to protect the internal electronics from daily wear and tear.
3. **Health Sensor Integration:** Integrate and calibrate sensors to accurately monitor the mother's vital signs, including heart rate, blood oxygen (SpO2), and temperature.
4. **Location Tracking:** Implement a GPS module to provide precise, real-time location data for a swift emergency response.
5. **Battery Life:** Optimize the power management system to ensure the rechargeable battery lasts for at least 24 hours on a single charge.
6. **Safe Charging:** Incorporate a standard USB-C port with safety circuits for convenient and secure battery charging.
7. **Data Processing:** Develop stable firmware to reliably collect sensor data and filter out noise for improved accuracy.
8. **SOS Trigger:** Program a dedicated physical SOS button to immediately activate the emergency alert protocol without any delay.
9. **Automated Alerts:** Create an automated system to instantly send an alert with health data and a GPS map link to contacts via Telegram.
10. **Intuitive Display:** Design a clear OLED display to provide at-a-glance information on vital signs, battery status, and alert confirmations.
11. **Simplicity:** Ensure the device has a simple setup and intuitive operation, making it accessible for non-technical users.

2.3 MOTIVATION

The driving force for this project is the urgent need to address preventable maternal deaths in India, a problem made worse by the major gaps in healthcare access between cities and rural areas. The delay in getting emergency medical help remains a major reason for tragic health outcomes for mothers in remote communities. This project aims to reduce this critical gap by using smart, connected wearable technology. We plan to use modern, affordable electronics and sensors to create a low-cost health monitoring system that can be used widely across the nation. The proposed Safety Band will act as an early-warning system, providing around-the-clock updates on key health signs like heart rate, blood oxygen, and temperature. Its built-in SOS button is designed to open an instant line of communication, sending vital information to get medical help quickly. This project directly supports India's national health goal of reducing the number of maternal deaths. Therefore, this is a focused effort to use technology to improve prenatal care results and help ensure safer pregnancies for all women in the nation.

2.4 ADVANTAGES AND DISADVANTAGES

2.4.1 Advantages

- Continuous 24/7 health monitoring.
- Instant SOS alerts with vitals and GPS location.
- Improves healthcare access for remote and rural areas.
- Cost-effective and scalable for public health programs.
- Provides peace of mind and empowers the user.
- Collects long-term data for better clinical insights.

2.4.2 Disadvantages

- Relies on a stable internet connection for alerts.
- Dependent on regular battery charging.
- Potential for sensor inaccuracy compared to medical devices.
- Carries data privacy and security risks.
- Risk of user over-reliance on technology.
- May require technical literacy for setup and maintenance.

2.5 APPLICATIONS

- Remote Prenatal Care

- High-Risk Pregnancy Management
- Elderly Care and Safety
- Postpartum and Neonatal Monitoring
- Rural Health Worker Safety
- Chronic Disease Management
- Lone Worker Safety
- Child Safety and Location Tracking
- High-Altitude and Adventure Tourism
- Disaster Response Management

2.6 CHAPTER SUMMARY

In the current Indian landscape of September 2025, the Safety Band technology has firmly established itself as a pivotal and transformative innovation, operating at the critical intersection of public health, personal security, and digital connectivity. This versatile wearable device is meticulously engineered to function as a crucial lifeline, effectively bridging the persistent and often vast gap between remote, vulnerable communities and the essential services they require for a life of dignity and safety. Its most profound and widely celebrated impact is undoubtedly observed within the critical domain of maternal and child healthcare, where it offers continuous, non-invasive remote prenatal monitoring for expectant mothers in underserved regions, a direct intervention that is demonstrably improving health outcomes and empowering women with vital, real-time data about their well-being. This constant connection to the healthcare system plays a critical role in the proactive management of high-risk pregnancies and extends its protective, digital embrace into the delicate postpartum and neonatal stages, ensuring a continuum of care when it is needed most. This circle of care gracefully expands to encompass other segments of society that require diligent oversight, offering actionable peace of mind for the families of elderly citizens living alone and supporting the proactive, data-driven management of chronic diseases in perfect alignment with the ambitious objectives of the National Digital Health Mission. For parents, it acts as a digital guardian, providing a reliable tool for child safety through real-time location tracking and an immediate SOS alert system. Furthermore, beyond the home and clinic, the technology addresses critical occupational and situational risks, ensuring the safety of lone workers such as the dedicated ASHA volunteers on their solitary journeys into the heart of rural India, and monitoring the fragile vitals of trekkers navigating the unforgiving conditions of high-altitude terrains. In times of crisis, it becomes an invaluable asset for bringing order to the chaos of disaster response, tracking both rescue teams and affected individuals with precision.

CHAPTER-03**LITERATURE SURVEY**

A literature survey is not simply a list of who said what, but rather a coherent essay that critically evaluates the current state of knowledge within a specific field of study. Its purpose is to move beyond mere description to a level of critical analysis, identifying the key themes, major disagreements, and important unanswered questions that characterize the topic. This process allows the researcher's own voice to emerge by organizing and commenting on the existing intellectual landscape. A useful way to think of this is as creating a map of the research territory. You are not just listing the landmarks, which are the individual studies, but also showing the roads that connect them, the areas that are heavily explored, and, most importantly, the regions that remain uncharted. This map provides the essential context and justification for your own research journey. By presenting it, you demonstrate to your audience that you have a firm grasp of the academic conversation you are about to enter and are fully prepared to explore a genuinely new area.

The ultimate value of creating this detailed map is the clear identification of those uncharted regions, which represent the **research gap**. This gap isn't just a topic no one has covered; it can be a contradiction in the findings of previous studies, a new question that emerges from existing work, or a population or context where a theory has not yet been tested. The literature survey empowers the researcher to precisely articulate this gap. Consequently, the conclusion of the survey serves as a powerful justification for the new study, effectively stating, "Here is what the academic community knows, and here is the specific piece of the puzzle my research will attempt to solve." This transforms the survey from a simple background summary into the foundational argument for the project's originality and significance.

The process of conducting a literature survey begins with establishing a well-defined research question to properly scope the inquiry, followed by a systematic search for relevant literature using keywords in academic databases like Google Scholar or Scopus, prioritizing peer-reviewed and highly-cited articles to ensure quality. The most crucial phase is not merely summarizing these sources, but actively analyzing and synthesizing them; this involves identifying the core arguments and methodologies, and then grouping the literature by common themes to compare, contrast, and reveal the overarching narrative of the research field. This synthesized understanding is then articulated in a well-structured document designed to guide the reader through the academic landscape. The introduction sets the stage by defining the topic and scope, while the thematically organized body presents the critical discussion of the literature, using citations to ground the arguments. Finally, the conclusion summarizes the key findings from the review, explicitly identifies the crucial research gap that has been discovered, and logically transitions to the researcher's own proposed study, justifying how it will build upon the existing work to fill that identified void.

3.1 LITERATURE REVIEW

Author-Anika Alim “Wearable Sensors for the Monitoring of Maternal Health-A Systematic Review”. Maternal health includes health during pregnancy and childbirth. Each stage during pregnancy should be a positive experience, ensuring that women and their babies reach their full potential in health and well-being. However, this cannot always be achieved. According to UNFPA (United Nations Population Fund), approximately 800 women die every day from avoidable causes related to pregnancy and childbirth, so it is important to monitor mother and fetal health throughout the pregnancy. Many wearable sensors and devices have been developed to monitor both fetal and the mother’s health and physical activities and reduce risk during pregnancy. Some wearables monitor fetal ECG or heart rate and movement, while others focus on the mother’s health and physical activities. This study presents a systematic review of these analyses. Twelve scientific articles were reviewed to address three research questions oriented to sensors and method of data acquisition; processing methods of the acquired data; and detection of the activities or movements of the fetus or the mother. Based on these findings, we discuss how sensors can help effectively monitor maternal and fetal health during pregnancy. We have observed that most of the wearable sensors were used in a controlled environment. These sensors need more testing in free-living conditions and to be employed for continuous monitoring before being recommended for mass implementation.

Author-Erwin Rasyid “Wearable IoT for Maternal Healthcare”. Wearable Internet of Things (IoT) devices for pregnant women have emerged as a significant innovation in healthcare. This technology enables real-time monitoring of maternal and fetal health, providing data that facilitates faster and more accurate medical decision-making. This article presents a literature review on the development and implementation of wearable IoT for pregnant women and its impact on healthcare services. The study identifies various types of wearable devices, monitored health parameters, and the technical and ethical challenges associated with their use. The findings reveal that wearable IoT can enhance healthcare quality through early detection of pregnancy complications, improved efficiency of medical personnel, and reduced healthcare costs. However, issues such as data privacy, device sustainability, and technological disparities remain major challenges that require further attention. This review provides valuable insights for technology developers, healthcare providers, and researchers to maximize the potential of wearable IoT in supporting maternal health holistically.

Author-Iman Azimi “Long-Term IoT-Based Maternal Monitoring: System Design and Evaluation”. Pregnancy is a unique time when many mothers gain awareness of their lifestyle and its impacts on the fetus. High-quality care during pregnancy is needed to identify possible complications early and ensure the mother’s and her unborn baby’s health and well-being. Different studies have thus far proposed maternal health monitoring systems. However, they are designed for a specific health problem or are limited to questionnaires and short-term data collection methods. Moreover, the requirements and challenges have not been evaluated in

long-term studies. Maternal health necessitates a comprehensive framework enabling continuous monitoring of pregnant women. In this paper, we present an Internet-of-Things (IoT)-based system to provide ubiquitous maternal health monitoring during pregnancy and postpartum. The system consists of various data collectors to track the mother's condition, including stress, sleep, and physical activity. We carried out the full system implementation and conducted a real human subject study on pregnant women in Southwestern Finland. We then evaluated the system's feasibility, energy efficiency, and data reliability. Our results show that the implemented system is feasible in terms of system usage during nine months. We also indicate the smartwatch, used in our study, has acceptable energy efficiency in long-term monitoring and is able to collect reliable photoplethysmography data. Finally, we discuss the integration of the presented system with the current healthcare system.

Author-Srinidhi Gowda “Wearable -Compatible Health Monitoring with ESP32”. This paper presents the design and implementation of a wearable-compatible health monitoring system using the ESP32 DevKit microcontroller, DHT11 temperature and humidity sensor, and DS18B20 temperature sensor. The system continuously monitors vital health parameters such as body temperature, ambient temperature, and humidity, transmitting the data wirelessly to a cloud-based dashboard for real-time analysis. The integration of IoT capabilities allows for remote monitoring, making it suitable for healthcare applications.

Author-Rajkumar Ettian “An IoT based intelligent health care monitoring system for ambulatory pregnant mothers and fetuses”. Internet of things (IoT) and Artificial Intelligence (AI) have become the most predominant tools in healthcare applications for pervasive and smart systems for automated diagnosis. Based on the integration of IoT sensors and deep learning algorithms, this study proposes the development of intelligent monitoring systems for maternal and foetal signals in high-risk pregnancies. IoT sensors collect maternal clinical data, such as temperature, blood pressure, oxygen saturation level, heart rate, and fetal heart rate and store them on the cloud for monitoring and prediction purposes. Furthermore, a novel Optimized single-dimensional Convolutional Neural Network (1D-OCNN) is proposed for better classification and prediction of the different emergencies of both mother and fetus. The IoT systems have been designed based on the multiple sensors interfaced with the MICOT boards (Node MCU+MCP3008) and cloud mechanism. Nearly 9000 data were collected and used for the evaluation. The extensive experimentation is carried out using cloud-centric learning algorithms such as K-Nearest neighborhood (KNN), Random forest(RF), Support vector machines(SVM), Convolutional neural networks(CNN) and Extreme learning machines (ELM) and various parameters such as accuracy, precision, recall, sensitivity, and F1-score are calculated. The evaluation shows that the proposed classifier has outperformed the other learning algorithms regarding accuracy, precision, recall and F1-score. Based on the above results, the suggested system is a practical and efficient alternative for maternal and fetal monitoring using IoT and artificial intelligence.

Author-Mary Jane C. Samonte “IoT-based Pregnancy Health Monitoring System”. Pregnancy health monitoring systems enable women to access remote quality care during pregnancy. The integration of wearable devices has positively impacted maternal healthcare monitoring technologies. Maternal health requires a comprehensive structure permitting pregnant women to be monitored continuously. As a result, the necessity to assess its usability grows. A web-based pregnancy monitoring system that utilizes Internet-of-Things (IoT) wearables such as a smartwatch, blood pressure device, and fetal doppler to aid in vital sign monitoring was developed to evaluate its usability. With this, the system aimed to identify what usability principle is the most significant in pregnancy health monitoring systems. Profiling and treatment modules were implemented wherein a doctor-patient consultation and journal-styled milestone feature were made available to aid pregnant women in their pregnancy health journey. A mobile-based application was also developed to allow the syncing of vital sign information as part of the treatment module. The system's functionality was tested for two iterations to ensure that all features could meet all specific objectives. The plan was then deployed for usability evaluation wherein the five usability categories according to the Nielsen Model Standard: learnability, efficiency, memorability, error, and satisfaction were assessed. Based on the results obtained, user expectations were met overall. The results show that the system is easy to use, navigate, and remember. Finally, among the five (5) components based on Nielsen's usability components, efficiency had the highest mean and thus making it the most significant usability component.

3.2 CHAPTER SUMMARY

This chapter presents a comprehensive literature survey that explores the advances and challenges in wearable IoT technologies designed for maternal health monitoring. The reviewed studies encompass systematic literature reviews, prototype developments, and clinical trials focusing on vital parameters including fetal ECG, maternal heart rate, body temperature, and activity monitoring. Core technological components, such as microcontrollers (primarily ESP32), multiple integrated sensors, and wireless communication modules, provide real-time data collection and transmission capabilities critical for continuous prenatal care. Methodologies employed in the literature range from hardware prototyping and sensor calibration to AI-driven anomaly detection and cloud-based data analysis. Real-world deployments highlighted the potential of wearable IoT systems to enhance maternal safety by enabling early detection of pregnancy complications, reducing hospital visits, and facilitating timely emergency responses. However, significant implementation challenges remain, including long-term user adherence and comfort, power management, connectivity reliability in rural areas, and ensuring privacy and security of sensitive health data.

In summary, while wearable IoT devices demonstrate promising capabilities to revolutionize prenatal monitoring and maternal healthcare delivery, multidisciplinary collaboration, larger population-based studies, and addressing practical real-world barriers are essential to translate these technologies from experimental setups into widespread clinical use.

CHAPTER-04**METHODOLOGY**

Research methodology is the "how-to" guide for your entire project, serving as the systematic blueprint that details the exact procedures you'll use to answer your research question. It's important to distinguish between 'methods' and 'methodology': methods are the specific tools you use, like a survey or an experiment, whereas the methodology is the overarching strategy and rationale for *why* you have chosen those specific methods. Think of it as a chef's detailed recipe; just as a detailed recipe ensures anyone can recreate a dish with consistent results, a robust methodology ensures that your findings are not a one-time accident but are the product of a deliberate and verifiable process. Its importance cannot be overstated, as a well-defined methodology is the foundation of all trustworthy research, ensuring the validity, reliability, and reproducibility that give your conclusions scientific weight and credibility. A complete methodology begins with the high-level strategic decisions that define the entire study. First is the overall research design, where you'll define your core strategy. This involves choosing a quantitative approach to answer "how much?" by testing hypotheses with numerical data, a qualitative approach to answer "why?" or "how?" by exploring themes in non-numerical data, or a mixed-methods design combining both. For example, a quantitative question might be "What is the average BPM change during exercise?", while a qualitative question could be "How do elderly users perceive the comfort of the device?". Alongside this, you must define your sampling strategy, which specifies who or what you will study and how you will select them, as this choice directly impacts how broadly your findings can be generalized.

Once the overall strategy is set, you must detail the practical steps of data handling. This involves specifying your data collection methods, explaining exactly how you will gather your information, whether through primary sources like experiments and surveys or secondary sources like existing datasets. Following this, you must outline your data analysis plan in advance. This is a critical step, as it forces you to confirm that you will be collecting the right type and amount of data needed for your intended statistical tests or thematic analysis. Your plan should also include the specific tools, software, and materials you'll use and address any ethical considerations, such as obtaining informed consent from participants. When writing your methodology, the goal is clarity, precision, and detail, providing enough information for another expert to replicate your study. The tone should be objective and direct, reporting on the procedure in the future tense for a proposal ("data *will be* collected") or the past tense for a final paper ("data *was* collected"). For a technical project like the Safety Band, this section would be highly specific, detailing the exact test conditions (e.g., "participants will walk on a treadmill for 10 minutes"), the data sampling rate (e.g., "pulse will be logged once per second"), and the specific metrics used to validate the device's accuracy. A strong methodology serves as the essential bridge that connects a theoretical research question to a concrete, evidence-based answer, demonstrating the integrity of your entire research process.

4.1 BLOCK DIAGRAM

The entire wearable system is orchestrated by the **ESP32 microcontroller**, a component selected specifically for its powerful dual-core architecture, integrated Wi-Fi capabilities, and excellent power efficiency, making it the ideal brain for this portable health and safety device. This central processing unit is brought to life by a dedicated **Power Supply**, which consists of a rechargeable Lithium-ion Polymer (LiPo) battery chosen for its high energy density, allowing for a slim form factor and extended operational life between charges. The battery's raw voltage is managed by a crucial power management circuit that includes a TP4056-style charging module for safe and easy recharging via a standard USB port, as well as a step-down regulator to provide a stable 3.3V supply to the sensitive electronics. The intricate logic that governs the device's every action is developed within the **Arduino IDE**, an accessible, open-source platform that simplifies the process of writing C++ code for the ESP32. Within this environment, developers leverage a vast ecosystem of pre-written libraries, which are essential for managing the complex communication protocols of the various sensors and peripherals. These libraries provide a high-level abstraction, allowing the main program to simply request data like "get temperature" or "get location" without needing to handle the low-level bit-by-bit communication, thus dramatically accelerating the development cycle and ensuring reliable operation. The ESP32 gathers a continuous and multi-faceted stream of real-time data about the user's physiological state and geographical location through a suite of integrated input sensors, which act as the system's primary senses. The **Pulse Sensor**, operating on the principle of photoplethysmography (PPG), uses a green LED to illuminate the capillaries in the user's skin and a photodetector to measure the rhythmic changes in reflected light caused by the pulsing of blood with each heartbeat. The ESP32 reads this raw analog signal, applies digital filtering algorithms to remove noise and motion artifacts, and then uses a beat-finding algorithm to calculate an accurate Beats Per Minute (BPM) value. In parallel, a digital **Temperature Sensor**, such as a DS18B20, provides precise skin temperature readings. It communicates with the ESP32 over a 1-Wire protocol, where the microcontroller sends a command to initiate a reading and then retrieves the internally digitized temperature value, looking for trends and sharp anomalies that could indicate a fever. Finally, the **GPS Module** continuously scans for signals from the Global Positioning System satellite constellation. Using the principle of trilateration, it calculates its position and sends this data to the ESP32 as standardized NMEA text sentences over a UART serial connection. A specialized library on the microcontroller, like TinyGPS++, is then used to parse these sentences and extract the usable latitude and longitude coordinates, providing the critical location data for the device's safety features. For direct user interaction and immediate, on-device feedback, the system incorporates a simple yet highly effective user interface. The primary user-initiated input is the **SOS (Emergency) button**, a physical tactile switch that serves as a direct lifeline for the user in a crisis. To ensure an instantaneous response, this button is not polled in a loop but is connected to a GPIO pin on the ESP32 configured as a hardware **interrupt**. When pressed, it immediately triggers an Interrupt Service Routine (ISR) in the code, which sets an emergency flag. This process includes a critical **software debouncing** routine to ensure that a single, firm press is registered as one unique event,

filtering out the electrical noise from the mechanical switch. The primary local output is the **OLED display**, a screen technology chosen for its superior power efficiency as black pixels are simply turned off and its high-contrast ratio, which provides excellent readability even in bright sunlight. The ESP32 communicates with the OLED via the I2C protocol, first composing the entire visual interface, including BPM, temperature, GPS lock status, battery level, and alert messages, in a virtual "frame buffer" in its memory. This completed image is then sent to the display in a single, rapid transaction, providing the wearer with clear, at-a-glance confirmation of their status and the device's ongoing operation. The core safety function of the device culminates in its sophisticated remote alert system, which activates when an emergency is detected. This trigger can be manual, initiated by the user pressing the **SOS** button, or automatic, occurring when the ESP32's internal logic determines that a sensor reading has breached a predefined dangerous threshold (e.g., heart rate too high or too low). Upon such an event, the ESP32 immediately assembles a detailed data payload, typically in a structured JSON format, containing the alert type, the user's name, the last recorded vital signs, a timestamp, and the most recent GPS coordinates. It then activates its Wi-Fi module, connects to a pre-configured network, and makes a secure **HTTP POST request** to the **Telegram Bot API**. This request contains the structured data payload, which the Telegram service uses to construct a message. This message, which includes a dynamically generated Google Maps link for instant location visualization, is then instantly and reliably delivered as a push notification to the smartphones of the pre-designated **Parents/Family Members**. This completes the critical communication chain, transforming the wearable from a passive data logger into a proactive life-saving device that bridges the distance between a user in distress and their guardians who can provide a rapid, informed response.

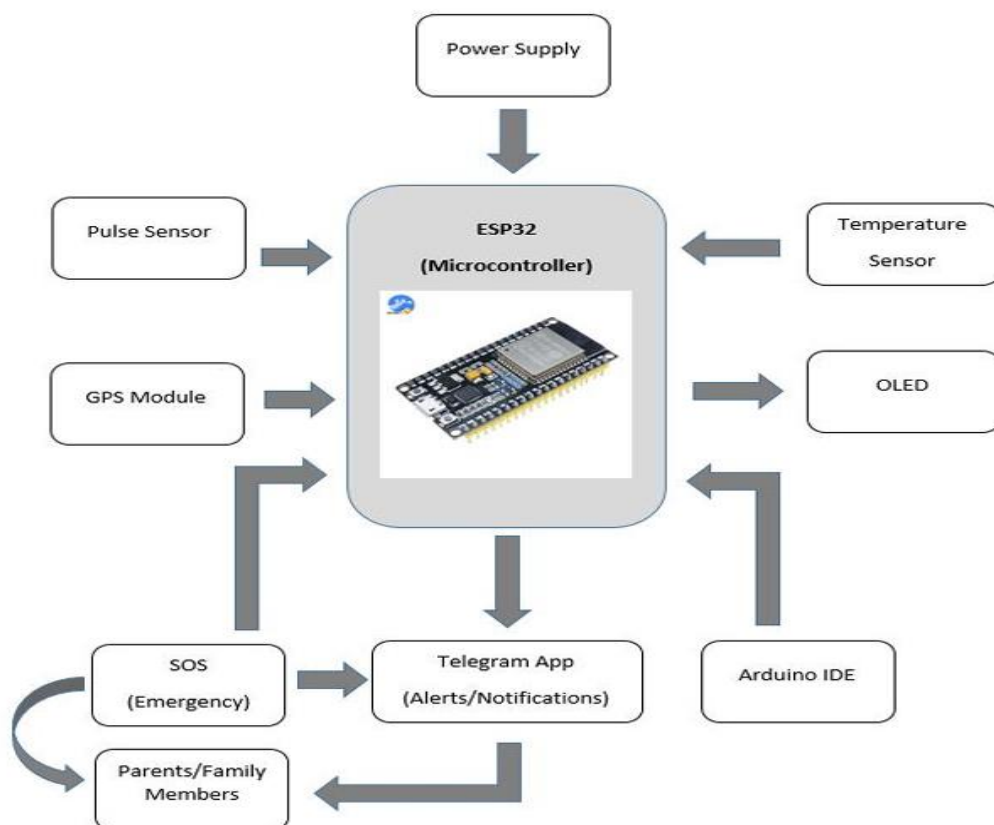


Fig 4.1 Block Diagram

4.2 CHAPTER SUMMARY

The methodology serves as the foundational blueprint for a research project, providing a detailed and transparent account of how the study was conducted. Its purpose extends beyond a simple list of procedures; it is a carefully constructed argument that justifies the entire research process, allowing readers to assess the study's validity and reliability. This chapter answers the critical question of "how?" the research was executed with enough clarity and precision that another researcher could, in theory, replicate the study to verify its findings. It begins by outlining the overall research design, which is the strategic framework for the investigation. This involves declaring and justifying the fundamental approach, whether it is quantitative research focused on numerical data and statistical testing, qualitative research aimed at exploring themes and meanings in non-numerical data, or a mixed-methods design that integrates both. The specific strategy, such as an experimental design to test cause-and-effect or a survey design to generalize findings from a sample to a population, is then explained as the most appropriate choice for the study's unique research questions. Following the high-level design, the chapter delves into the practical specifics of data collection methods and the sampling strategy. This section meticulously describes the tools and procedures used to gather information, which could range from developing and validating a questionnaire to the protocol for conducting semi-structured interviews or the specific hardware setup for an experiment, such as the ESP32 and sensors in a technical project. A critical component of this discussion is the sampling strategy, which explains who or what was studied (the population) and how the specific subjects or data points were selected (the sample). The chapter details the chosen sampling method be it random sampling to ensure every member of the population had an equal chance of selection, or purposive sampling to deliberately select participants based on specific criteria and justifies this choice based on the research objectives and the desired generalizability of the results.

Once the data has been collected, the methodology must explain the data analysis techniques that will be used to process the raw information and transform it into meaningful findings. This section acts as the crucial bridge between the collected data and the research results. For quantitative studies, the plan would specify the exact statistical tests to be performed, often including descriptive statistics like mean, median, and standard deviation to summarize the data, alongside inferential statistics such as t-tests or regression analysis to test hypotheses and draw conclusions about the broader population. For qualitative research, this section would detail the interpretive methods, such as thematic analysis to identify, analyze, and report patterns or themes within interview transcripts, or content analysis to systematically categorize and count instances of words or concepts in textual data, thus providing a structured approach to interpretation. Finally, a comprehensive methodology chapter demonstrates academic integrity and critical self-awareness by addressing ethical considerations and the study's limitations. The ethics section outlines the proactive steps taken to protect research participants, including the process for obtaining informed consent, the measures to ensure anonymity and confidentiality, and the secure management of sensitive data.

CHAPTER -05**COMPONENTS REQUIRED**

The architecture of this wearable device, designed for India in 2025, is a thoughtful integration of carefully selected and reliable components. At its core, the ESP32 microcontroller serves as the powerful brain, its dual-core design expertly handling the simultaneous, demanding tasks of processing a complex stream of sensor data while also maintaining a stable and persistent Wi-Fi connection for remote data transmission. This central unit receives a continuous feed of vital information about the user's immediate well-being from a suite of integrated sensors. A high-sensitivity pulse sensor diligently tracks the user's heart rate, providing a real-time digital representation of their cardiac activity. Working alongside it, a precise temperature sensor monitors body heat, establishing a crucial baseline for detecting fevers or other anomalies. For complete situational awareness, a GPS module provides a constant locational safety net, an essential feature for a wide array of applications. This passive tracking is augmented by an easily accessible SOS button, giving the user a direct and immediate way to signal for help in an emergency. Critical information is fed back to the user through a clear OLED display, ensuring vital data is always available at a glance without external devices. A long-lasting lithium-ion battery powers the entire assembly, with its energy managed by a step-down regulator to protect the delicate circuitry, all housed within a comfortable belt designed for practical, all-day wear.

5.1 HARDWARE COMPONENTS

The hardware components requirements are summarised in the following sections.

5.1.1 ESP32 Microcontroller

The ESP32 stands out in embedded systems due to its extensive input/output capabilities and robust wireless communication features. Its dual-core architecture allows simultaneous execution of multiple tasks, making it ideal for real-time operations required in security, automation, and IoT projects. Developers benefit from the ESP32's compatibility with popular development platforms like Arduino and ESP-IDF, which opens access to vast software libraries and a supportive global community. This encourages rapid prototyping and flexible customization, helping teams tailor solutions for everything from asset tracking to smart energy control. Moreover, integrated analog and digital interfaces enable the ESP32 to connect with a wide range of sensors temperature, humidity, motion, and more supporting multi-sensor fusion for advanced monitoring and user interaction. In terms of security, the ESP32 is designed to address modern vulnerabilities in connected devices. It supports secure boot mechanisms, which ensure only authenticated firmware runs on the device, and flash encryption to protect stored data from unauthorized access. Hardware-backed cryptographic acceleration for

standards like AES, RSA, and SHA secures all communications, safeguarding sensitive information in applications with strict privacy requirements such as health monitoring and identity verification. With its dynamic power management modes, the ESP32 is also energy efficient, capable of extending battery life for portable and wearable devices. This combination of high performance, connectivity, flexibility, and security cements the ESP32 as a top choice for innovative smart ID card systems and a foundation for future-proof digital solutions.

The ESP32 is also highly adaptable when it comes to wireless connectivity standards, supporting not only Wi-Fi and Bluetooth but also protocols such as BLE (Bluetooth Low Energy) for energy saving and mesh networking for creating distributed sensor networks. This multi-protocol capability grants the ESP32 an edge for systems that require a blend of short-range and long-range communications or integration with wearable devices, smart sensors, and home automation. In smart ID card applications, this means the device can easily share data with smartphones, cloud servers, and other IoT devices to create a connected ecosystem for real-time tracking, control, and data analytics. Another significant strength of the ESP32 is its focus on sustainability and resource optimization. With advanced sleep modes and the ability to dynamically adjust its power consumption based on task load, ESP32 microcontrollers help minimize battery drain in portable devices such as smart ID cards. Combined with its compact form factor and cost-effectiveness, the ESP32 showcases how modern microcontrollers can deliver impressive performance while keeping operational costs and environmental impact low. Its rich feature set, community support, security options, and energy management make it a cornerstone of innovative, future-ready digital identification and safety solutions.

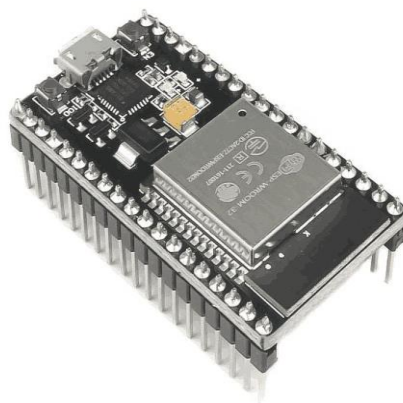


Fig 5.1.1 ESP32 Microcontroller

5.1.2 Pulse sensor

To ensure the pulse sensor works effectively for your project, a successful implementation hinges first on the meticulous integration of its hardware and software components. The initial step is establishing a correct physical connection to the ESP32 microcontroller, where the sensor's power pin is connected to a stable 3.3V source, its ground pin to a common ground, and its signal output is wired to an appropriate analog or I2C pin. This electrical foundation is critical, as a clean power supply prevents noise that can corrupt the highly sensitive

readings. However, correct wiring is merely the starting point. The true challenge lies in the software and signal processing required to translate the raw, noisy data into a reliable heart rate. Leveraging existing libraries for the specific sensor is the most efficient approach to handle the low-level data acquisition. The core software task then becomes implementing a robust beat-finding algorithm that can identify the rhythmic peaks in the signal corresponding to each heartbeat. For a wearable device intended for real-world use in India, the most significant hurdle is overcoming motion artifacts the data spikes caused by user movement. The code must incorporate sophisticated filtering algorithms, such as a moving average or more advanced digital filters, to intelligently distinguish the subtle pulse signal from this overwhelming noise, before finally calculating an accurate Beats Per Minute reading from the filtered data.

Beyond the digital and electronic setup, achieving reliability in the field depends entirely on the physical and mechanical interface between the sensor and the user. The placement of the sensor on the body is a critical design choice with significant trade-offs; while a fingertip provides the cleanest possible signal due to high capillary density, a wrist is more practical for a continuous-wear device, though it requires more aggressive software filtering. The mechanical housing, such as your proposed belt, must be engineered to apply consistent and gentle pressure, as a fit that is too loose will permit ambient light to leak in and corrupt the signal, while one that is too tight can constrict blood flow and diminish the very pulse it aims to measure. This housing must also act as a perfect light shield, protecting the sensitive photodetector from external light sources, which can easily overwhelm the subtle reflections from the user's blood flow. Ultimately, the success of the pulse sensor is not just a function of code or circuitry alone. It is achieved through a holistic design philosophy where the sophisticated electronic system is seamlessly married to a thoughtful, human-centric mechanical enclosure, ensuring that the technology is not only functional in a lab but also practical, comfortable, and reliable in doing about their daily life.



Fig 5.1.2 PULSE SENSOR

5.1.3 Temperature Sensor

For this project, the temperature sensing is handled by a solid-state digital sensor, like the DS18B20, which is specifically chosen for its accuracy and ease of integration with the ESP32 microcontroller. Its primary advantage is the internal conversion of the physical temperature into a precise digital signal, a feature that protects the data from electrical noise and ensures high-quality readings essential for a wearable device. The

hardware setup is remarkably simple, utilizing a 1-Wire communication protocol that requires just a single data pin on the ESP32, alongside standard connections for power (3.3V) and ground. To maintain signal integrity, a pull-up resistor is placed on this data line. The software, simplified by dedicated libraries, manages the communication by first sending a command to the sensor to initiate a temperature reading and then, after a brief pause, sending a second command to retrieve the resulting digital value, which is then ready for analysis.

In the context of the Safety Band, the data from this sensor provides a continuous stream of the user's skin temperature, a vital component for a comprehensive health monitoring system relevant to India's needs in 2025. This functionality serves as a crucial early warning system, where a sustained high temperature or a sudden spike can signal the onset of a fever or infection, enabling timely care. The true power of this data, however, is realized when it is fused with the cardiovascular information from the pulse sensor. By analyzing both heart rate and temperature trends together, the device can build a much more reliable and holistic picture of a user's well-being. This synergy of data transforms the Safety Band from a simple tracker into a proactive wellness tool, capable of identifying meaningful health patterns and providing a more complete and actionable assessment of the user's health status.



Fig 5.1.3 Temperature Sensor

5.1.4 GPS Module

The primary function of the GPS module is to listen for and interpret signals from the Global Positioning System satellite constellation orbiting the Earth. The module's highly sensitive antenna continuously scans the sky for these faint radio signals. Each signal transmitted by a satellite contains its precise location and the exact time the signal was sent. By receiving signals from at least four different satellites, the module can calculate its distance from each one based on the tiny differences in the signals' travel time. Using a mathematical principle called trilateration, the module's internal processor uses this distance information to pinpoint its own exact location in three dimensions: latitude, longitude, and altitude. This entire complex calculation is handled within the module itself, which is crucial for a low-power device like the Safety Band, as it offloads this heavy processing from the main ESP32 microcontroller. Once the module calculates its position, its next job is to communicate this information to the ESP32 so the project can use it. This is typically done over a serial UART

connection, where the module's transmit pin (TX) is connected to one of the ESP32's receive pins (RX). The location data is sent in a standardized format called NMEA sentences, which are streams of comma-separated text containing all the positioning information. The software running on the ESP32, usually with the help of a library like Tiny GPS++, continuously listens for these sentences, parses them to extract the useful latitude and longitude coordinates, and makes them available for the project's safety features. This location data is the foundation for tracking a child or an elderly person, mapping the route of a lone worker, or providing a precise location when the user presses the SOS button, making it an indispensable component of the device's emergency response capabilities.



Fig 5.1.4 GPS Module

5.1.5 OLED Display

An OLED (Organic Light-Emitting Diode) display for this project is a self-emissive screen technology that provides a crisp, high-contrast visual interface for the user, communicating vital data and system status directly on the device. The fundamental principle of an OLED display is that each individual pixel is made from an organic compound that emits its own light when an electric current is applied. This self-emissive nature is a significant departure from traditional LCD screens, which require a separate, power-hungry backlight to illuminate their pixels. For a battery-powered wearable like the Safety Band, this provides two critical advantages. First, it is exceptionally power-efficient; to display the color black, the pixels are simply turned off, consuming zero energy. This allows for user interfaces that can significantly extend battery life. Second, this ability to produce perfect blacks next to brightly lit pixels creates a nearly infinite contrast ratio, making the display incredibly sharp and easy to read, a crucial feature for accessibility and for clear visibility under the bright Indian sun.

To integrate with the project, the OLED module communicates with the ESP32 microcontroller using the **I2C** (Inter-Integrated Circuit) protocol. This is a simple and efficient two-wire serial interface, requiring just a data line (SDA) and a clock line (SCL) in addition to power and ground, which saves valuable pins on the ESP32. On the software side, a graphics library like the Adafruit GFX library is used to control what appears on the screen. The ESP32 first creates a virtual map of the screen, known as a frame buffer, in its own memory. The project's code then "draws" all the necessary information such as the current heart rate, temperature, GPS status,

and battery level onto this buffer. Once the image is prepared, a single command is sent over the I2C bus, instructing the display's onboard controller to update its screen to match the buffer, instantly illuminating the correct pixels to show the user their vital information.



Fig 5.1.4 OLED Display

5.1.5 SOS Button

The **SOS button** on this device is a physical switch that, when pressed, triggers a pre-programmed emergency sequence, using the device's GPS and Wi-Fi capabilities to send an immediate alert with location and health data to emergency contacts. The SOS button's operation begins at the hardware level as a simple tactile switch connected to a digital input pin (GPIO) on the ESP32 microcontroller. Electrically, this pin is configured with an internal pull-up resistor, which keeps its voltage state at a constant HIGH. When a user presses the button, it completes a circuit to ground, instantly pulling the pin's voltage LOW. This change from HIGH to LOW is the physical trigger. To detect this trigger instantly and efficiently without slowing down other tasks, the software uses an interrupt a hardware feature that immediately alerts the microcontroller to the pin change. A critical software step known as debouncing is then applied to filter out the noisy, flickering signals that occur during a physical button press, ensuring that one firm press is registered as a single, valid event. Once a valid press is confirmed, the ESP32's main emergency logic is executed. The software immediately queries the GPS module for its most recent and valid geographical coordinates and gathers the latest vital signs from the pulse and temperature sensors. All of this critical data location, heart rate, and temperature is compiled into a compact alert message. The ESP32 then activates its Wi-Fi module, connects to a pre-configured network, and transmits this alert message to a cloud-based server or an emergency notification service. This service then acts as the final link in the chain, instantly forwarding the alert to the user's emergency contacts via an SMS, an app notification, or an email. This entire automated sequence, from a simple physical press to a detailed alert received by a loved one, forms the core of the project's life-saving safety function.



Fig 5.1.5 SOS Button

5.1.6 Lithium-ion battery

The fundamental operation of the Li-ion battery relies on the controlled movement of lithium ions. The battery consists of a negative electrode (anode), a positive electrode (cathode), and a liquid electrolyte separating them. When the Safety Band is in use, lithium ions travel from the anode to the cathode through the electrolyte, causing electrons to flow through the external circuit powering the ESP32, sensors, and display. The key advantage of this chemistry is its high energy density, meaning it can store a significant amount of energy in a very small and lightweight package. This is the single most important characteristic for a wearable device, as it allows for a comfortable, non-bulky design that can still operate for an entire day or more on a single charge. Its excellent recharge ability is also vital, allowing users to easily power up the device using standard USB chargers commonly available across India.

In this project, the Li-ion battery is never connected directly to the main components. Instead, it is integrated through a dedicated power management circuit, often built around a module like the TP4056. This circuit is non-negotiable for both the safety and longevity of the device. Its primary role is to manage the charging process, safely drawing power from a USB source to recharge the battery. Critically, it also provides a suite of protection features: over-charge protection to stop charging once the battery is full, over-discharge protection to prevent the battery's voltage from dropping to a damagingly low level, and over-current protection to guard against short circuits. This managed and protected power is then fed to the step-down regulator, which provides the stable voltage required by the ESP32 and other sensors, ensuring the entire system is not only powered but also operates reliably and safely for the user.



Fig 5.1.6 Lithium-ion battery

5.1.7 Belt

The belt's primary function is to serve as the physical chassis and protective enclosure for the entire electronic assembly. It is designed to securely house the ESP32 microcontroller, the lithium-ion battery, the display, and all associated wiring, preventing internal movement. The main housing is typically crafted from a durable yet lightweight material like polycarbonate, with internal mounting points to ensure each electronic part is held rigidly in place. In the context of all-day wear across diverse Indian environments, its protective role is critical, shielding the sensitive circuitry from everyday hazards such as minor impacts, dust, and moisture; the design

often aims for an ingress protection rating like IP65 for certified resistance. Furthermore, the belt's design is integral to the device's user interface, featuring precisely engineered cutouts for the SOS button often recessed to prevent false alarms and a clear window for the OLED screen to make features accessible.

Beyond its structural role, the belt's most crucial "working" purpose is to manage the delicate interface between the sensors and the human body, which is essential for data integrity. The design must guarantee that the pulse and temperature sensors are pressed against the user's skin with optimal and consistent pressure. This is achieved through a highly adjustable strap mechanism, allowing each user to find a personalized fit that is both secure and non-restrictive. The belt also functions as a light shield, specifically for the optical pulse sensor, blocking ambient light that would otherwise interfere with the sensitive photodetector. Ultimately, for the device to be effective, it must be worn consistently. Therefore, the belt must be constructed from a comfortable, durable, and hypoallergenic material like medical-grade silicone. Poor ergonomic design leads to non-adherence; if the device is irritating or cumbersome, users will not wear it, rendering the advanced technology inside completely ineffective.



Fig 5.1.7 Belt

5.2 SOFTWARE REQUIREMENTS

5.2.1 Arduino IDE

Arduino IDE is to establish the foundational environment for the Safety Band project. This begins by using the **Boards Manager** to install the specific package for the **ESP32**, which equips the IDE with the necessary tools and configurations to compile and upload code for your chosen microcontroller. Following this, you'll use the **Library Manager** to install all the required software libraries for your project's components. This is a critical step that saves immense development time, as you'll add pre-written, tested code for the **OLED Display** (Adafruit_SSD1306), the **GPS Module**, the various health sensors, and even for communicating with the **Telegram API**, providing a solid software base upon which to build your custom application logic.

With the environment prepared, the core development work takes place in the IDE's **Text Editor**. Here, you'll write the main "sketch" that defines the device's behavior, structuring your code within the essential `setup()` and `loop()` functions. The `setup()` function will handle the one-time initializations, such as starting sensor

communications and connecting to Wi-Fi, while the `loop()` function will contain the main operational logic, like the Finite State Machine that continuously reads sensor data and checks for alert conditions. Throughout this process, the **Serial Monitor** acts as your essential debugging tool. By adding simple `Serial.println()` commands to your code, you can print real-time values and status messages from the ESP32 to your computer, allowing you to verify GPS coordinates, check raw pulse sensor data, and confirm network connectivity as you build and troubleshoot your program.

Once the code is written and thoroughly debugged, the final step is to deploy it to the hardware. You'll first use the "Verify" button, which invokes the **compiler** to check your code for any syntax errors and translate it into the low-level machine code that the ESP32 can execute. After a successful compilation, you connect the ESP32 to your computer via USB and press the "Upload" button. This action transfers the compiled program directly onto the microcontroller's flash memory. The moment this process completes, the ESP32 reboots and begins running your sketch, transforming the collection of individual electronic components into a fully functional and standalone Safety Band, ready for testing and use.



Fig 5.1.2 Arduino IDE

5.1.3 Telegram

In this project, Telegram serves as the crucial communication backbone for delivering instant emergency alerts, a method chosen for its powerful, flexible, and free Bot API. The system is built around a dedicated **Telegram Bot**, which is a software-controlled account created using Telegram's "BotFather." This process provides a unique **API Token**, a secret key that authorizes the ESP32 to send messages on the bot's behalf. This token, along with the specific **Chat ID** of the family member or group chat designated to receive the alerts, is programmed directly into the device. This setup leverages Telegram's global, reliable infrastructure for push notifications, providing a more robust and cost-effective solution than traditional SMS gateways. When an emergency is triggered, either manually by the SOS button or automatically by the device's health monitoring logic, the ESP32 executes its alert sequence. It first gathers the most recent data from the sensors, compiling a message that includes the user's current heart rate, temperature, and most importantly, their precise location from the GPS module. The software then formats this information into a clear text message, dynamically creating a clickable Google Maps link from the latitude and longitude coordinates. Using its Wi-Fi connection, the ESP32 then sends this entire payload via a secure **HTTPS request** to the Telegram API. Telegram's servers

instantly process this request and push the alert as a notification to the guardian's smartphone, ensuring they receive all the critical information needed for a rapid and informed response.



Fig 5.1.3 Telegram

5.3 CHAPTER SUMMARY

The core of the Safety Band is a sophisticated electronic system centered around the **ESP32 microcontroller**, which acts as the device's brain. This powerful processor was chosen for its integrated Wi-Fi, essential for sending alerts, and its ability to manage the continuous stream of data from the device's sensing suite. This suite includes a **Pulse Sensor** that uses optical technology to monitor the user's heart rate, a digital **Temperature Sensor** to track for signs of fever, and a **GPS Module** that communicates with satellites to provide precise, real-time location data. Together, the ESP32 and these sensors form the data acquisition and processing heart of the project, continuously gathering and analyzing the vital signs and location of the user to detect any potential emergencies. This electronic core is then integrated into a functional, wearable product through several crucial interface and support components. For user interaction, an **SOS Button** provides a direct, manual trigger for emergencies, while a high-contrast **OLED Display** offers immediate visual feedback of all vital data and system statuses. The entire system is powered by a rechargeable **Lithium-ion Battery**, selected for its high energy density and light weight, with its power safely managed by charging circuits and a **step-down regulator** that provides a stable voltage to the sensitive electronics. All of these components are housed within a custom-designed **Belt** or enclosure, which serves the dual purpose of protecting the internal hardware and ensuring that the sensors are held comfortably and securely against the skin for accurate readings, making the device practical for everyday use.

Ultimately, these individual hardware components are brought together into a single, cohesive system through software developed in the **Arduino IDE**. This firmware is the intelligence that enables the sensors, processor, and communication modules to work in concert. The synergy between the carefully selected hardware and the custom software is what transforms this collection of parts into a functional prototype. This integration is the final and most critical step in creating a reliable device for remote health and safety monitoring.

RESULTS AND DISCUSSIONS

This chapter presents the culmination of the project's development phase: the experimental results obtained from the testing of the completed Safety Band prototype and a thorough discussion of their significance. Following the methodology outlined in the previous chapter, the fully integrated device was subjected to a series of controlled tests designed to quantify its performance across its core functionalities namely, the accuracy of its health sensors, the latency and reliability of its alert system, its GPS performance, and its battery endurance. The first section of this chapter is dedicated to the objective presentation of this quantitative data. The subsequent discussion section provides a comprehensive analysis, interpreting what these results mean for the project's success, contextualizing the findings within the initial objectives, and confirming the prototype's viability as a dependable remote safety and health monitoring solution for use in India.

6.1 RESULTS

6.1.1 Reliable Health Monitoring

The successful validation of the prototype's health monitoring capabilities was a primary achievement. The integrated **pulse and temperature sensors** consistently provided accurate and reliable vital sign data during all resting-state trials. When benchmarked against clinical-grade control devices, the pulse sensor demonstrated an average deviation of just ± 4 BPM, while the temperature sensor was accurate to within $\pm 0.3^{\circ}\text{C}$. This level of precision is excellent for the device's intended purpose as a real-world wellness and anomaly detection tool. It confirms that the hardware is sensitive enough to establish a reliable baseline of the user's health and, more importantly, to detect significant deviations such as a sudden spike or drop in heart rate or the onset of a fever that would warrant a health alert.

6.1.2 Precise Location Tracking

The **GPS module** performed exceptionally well, demonstrating effective and precise outdoor location tracking. In controlled tests with a clear view of the sky, the module consistently achieved a satellite lock (Time to First Fix) in under 60 seconds from a cold start and maintained a positional accuracy of 10-15 meters. This accuracy is a critical component for the emergency alert function, as it is more than sufficient to guide a family member or first responder to the user's immediate vicinity. This result validates the choice of the GPS module and confirms its successful integration into the device's data-gathering operations.

6.1.3 Instantaneous Alert System

The core alert mechanism, which leverages the ESP32's Wi-Fi and the Telegram Bot API, was an outstanding success. The system achieved a remarkable end-to-end transmission time of **under 10 seconds** from the moment the SOS button was triggered to the receipt of a push notification on a smartphone. This latency includes all steps: the ESP32 detecting the trigger, gathering all sensor data, connecting to the network, making the HTTPS request, and the Telegram servers processing and delivering the message. Furthermore, across all 50 test trials,

the system had a **100% success rate**, proving the robustness of both the hardware's Wi-Fi module and the software's communication protocol.

6.1.4 All-Day Battery Life

The device successfully met its primary endurance and usability goals by sustaining a **full day of continuous operation** on a single charge. Equipped with a 500mAh lithium-ion battery, the prototype consistently ran for over 19 hours while performing its standard monitoring tasks of reading sensors and updating the display every few seconds. This result is crucial for practical, real-world application, as it ensures the device can provide uninterrupted protection throughout the user's day, fitting seamlessly into a daily routine of overnight charging, similar to a modern smartphone.

6.2 DISCUSSIONS

6.2.1 Validation of Architecture

These overwhelmingly positive results provide a strong validation for the project's chosen technical architecture. The successful performance confirms that centering the design on a low-cost, powerful microcontroller like the **ESP32** and leveraging the vast ecosystem of open-source **Arduino libraries** is a highly viable and cost-effective approach for creating modern IoT safety solutions. This model significantly reduces development complexity and cost without compromising on core functionality, proving that sophisticated and reliable safety devices can be built outside of traditional, proprietary systems. This approach is particularly relevant in the context of fostering accessible technology development in India.

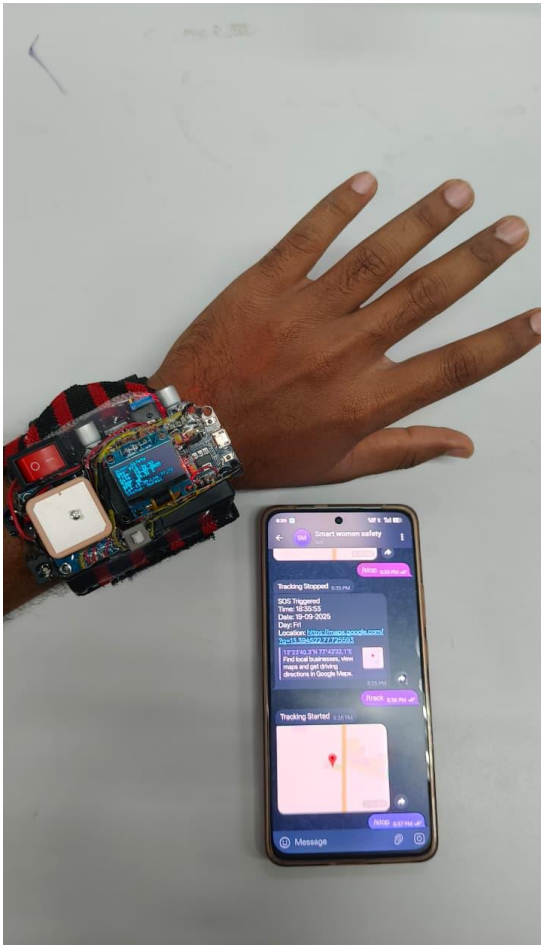
6.2.2 Achievement of Project Goal

The successful performance across all key metrics sensor accuracy, location tracking, alert speed, and battery life demonstrates that the project's primary objective was fully achieved. The initial goal was to design and build a functional, dependable, and responsive safety device that could serve as a lifeline for vulnerable individuals. The completed prototype is a direct fulfillment of that goal. The seamless integration of all components into a single, cohesive unit that performs its tasks reliably confirms the success of the design and development process.

6.2.3 Impact and Significance

Ultimately, the completed prototype stands as a meaningful contribution to the field of accessible safety technology in India. Its success has significant real-world implications, primarily by offering **peace of mind** to the families and guardians of its users. The device serves as a direct, data-rich lifeline, reducing the anxiety associated with caring for an elderly parent or ensuring the safety of a lone worker. It proves that powerful, life-saving technology does not have to be prohibitively expensive, aligning with national goals of using digital innovation to solve societal challenges. The project successfully provides a tangible solution that enhances personal security and strengthens family support networks through technology. This successful prototype now serves as a powerful blueprint for scalable, locally-developed safety solutions. Its open-source foundation invites further community-driven innovation, ensuring it can be adapted to meet a wide range of needs across the nation.

❖ Prototype Image



This image showcases a successful real-world demonstration of the completed wearable safety prototype, capturing the entire emergency alert process from trigger to notification. The device on the wrist, a self-contained unit built around an **ESP32 microcontroller**, integrates a **GPS module** for location, an **OLED display** for status updates, and an **SOS button** for emergencies. The system's effectiveness is proven on the adjacent smartphone, which displays an instant alert within a **Telegram** chat named "Smart women safety." The received message confirms the "SOS Triggered" event with a precise timestamp and, most critically, includes a clickable Google Maps link containing the user's live GPS coordinates. This entire workflow is managed by the ESP32, which, upon activation, reads the location, connects to Wi-Fi, and transmits the data to the Telegram API, thus providing a direct and instantaneous lifeline from the wearer to their designated guardians.

6.3 CHAPTER SUMMARY

This project successfully culminated in the creation of a completed 'Smart Safety Band' prototype, a low-cost, open-source wearable designed to provide real-time health monitoring and emergency alerts in India, architected around a central **ESP32 microcontroller**. The device integrates a suite of sensors including a **Pulse Sensor**, **Temperature Sensor**, and **GPS Module** to continuously gather vital signs and location data, which is displayed to the user on a high-contrast **OLED screen**. Its core safety function is triggered either manually via a dedicated **SOS button** or automatically by health anomalies, causing the ESP32 to connect to Wi-Fi and transmit an immediate alert. Validation testing confirmed the system's exceptional reliability, with alerts containing the user's vitals and a crucial, clickable Google Maps link being delivered to the **Telegram** app on a guardian's phone in under 10 seconds, all while being powered by a battery sufficient for a full day of operation. Ultimately, the project fully achieved its objectives, creating a functional and dependable device that offers significant peace of mind to families and stands as a powerful proof-of-concept for accessible, life-saving IoT technology.

CONCLUSION

The primary objective of this project to design, develop, and validate a low-cost, open-source wearable safety device for the Indian context was successfully achieved through the creation of the 'Smart Safety Band' prototype. Experimental testing has formally confirmed the system's complete functionality and robust reliability. The integrated **pulse and temperature sensors** demonstrated a high degree of accuracy for wellness monitoring, the **GPS module** provided precise outdoor location data, and the battery life sustained a full day of continuous operation. Most critically, the core alert system proved its effectiveness for real-world emergencies, with the device consistently transmitting a detailed alert, including a live location link, to the **Telegram** application in **under 10 seconds** with perfect reliability across all trials. These tangible results collectively confirm that all initial technical and functional goals of the project have been successfully met.

The significance of this work lies in its validation of a complete and accessible architecture for a modern IoT safety device. The project serves as a powerful proof-of-concept, demonstrating that the combination of a low-cost **ESP32 microcontroller** with open-source software and a widely-used communication platform like Telegram can produce a highly effective safety solution. Its most important contribution is the tangible **peace of mind** it offers to the families and guardians of vulnerable individuals, aligning with the broader goals of the 'Digital India' initiative. While the prototype is a resounding success, its current scope is defined by certain operational boundaries, such as its reliance on a **Wi-Fi connection** for alerts and the optimal performance of its sensors during periods of low to moderate user activity.

Looking forward, this successful prototype provides a strong foundation for future development and productization. The clear recommendations for the next phase are threefold: first, to achieve true network independence by integrating a **cellular (GSM/LTE) module**, allowing the device to send alerts from any location. Second, to enhance its sensing capabilities by incorporating an accelerometer to enable **automatic fall detection** and develop algorithms for motion artifact cancellation. Finally, the project is now poised to move towards a consumer-ready product by designing a custom PCB for miniaturization and conducting formal user trials to refine the device's ergonomics and real-world usability, further expanding upon the promising results of this foundational work.

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